

Thermodynamics Syllabus

Chapter 1: Basic Concepts & Definitions

Topics

- What is thermodynamics about?
- System and surroundings
- Extensive and intensive properties
- State, process and cycle

Learning Outcomes

- Define system, surroundings, closed / open / isolated systems.
- Distinguish between extensive and intensive properties with examples.
- Explain the concepts of state, process, cycle and equilibrium in thermodynamics.
- Describe engineering applications of thermodynamics (e.g., power plants, refrigeration) and identify system boundaries.

Suggested textbook pages: Chapter 1 (approx. pages 1-25)

Chapter 2: Thermodynamic Properties of a Pure Substance

Topics

- Pure substance definition
- Thermodynamic properties (u , h , s , v etc.)
- Phase diagrams (T - v , P - v , T - s)
- Thermodynamic tables

Learning Outcomes

- Describe what a pure substance is and its relevance in engineering thermodynamics.
- Interpret phase diagrams and locate states of a pure substance.
- Use thermodynamic property tables to determine internal energy, enthalpy and entropy of a given state.
- Explain the concepts of critical point, triple point and quality of a two-phase mixture.

Suggested textbook pages: Chapter 2 (approx. pages 26-60)

Chapter 3: Ideal and Real Gases

Topics

- Ideal gas and the ideal gas equation of state
- Real gas behaviour and compressibility factor
- Applications of ideal and real gas models

Learning Outcomes

- State the ideal gas equation of state and identify conditions when it is valid.
- Explain deviation from ideal behaviour and define the compressibility factor.
- Apply real gas models to engineering problems where ideal gas law fails.

Suggested textbook pages: Chapter 3 (approx. pages 61-85)

Chapter 4: The First Law of Thermodynamics for Closed Systems

Topics

- Internal energy in a system
- Heat transfer across a boundary
- Work (boundary work, shaft work etc.)
- First law for closed systems

Learning Outcomes

- Explain internal energy and its role in closed-system energy analysis.
- Distinguish heat transfer and work interactions across system boundaries.
- Formulate and apply the first law of thermodynamics for closed systems to real-life engineering devices.

Suggested textbook pages: Chapter 4 (approx. pages 86-120)

Chapter 5: The First Law of Thermodynamics for a Control Volume

Topics

- Enthalpy and flow work

- Mass and energy conservation equations in a control volume
- Applications in steady-flow devices (turbines, compressors, heat exchangers)

Learning Outcomes

- Define enthalpy and explain why it is used in control-volume analyses.
- Apply the mass and energy conservation equations for a control volume under steady flow conditions.
- Analyse key engineering devices (e.g., turbines, compressors, heat exchangers) using the first law.

Suggested textbook pages: Chapter 5 (approx. pages 121-155)

Chapter 6: Entropy and the Second Law of Thermodynamics

Topics

- Heat engines
- Refrigerators & heat pumps
- Second law: Kelvin-Planck and Clausius statements
- Carnot cycles
- Entropy and entropy generation
- Second law for closed and open systems

Learning Outcomes

- State and explain the Kelvin-Planck and Clausius statements of the second law.
- Define entropy, entropy generation and comment on irreversibility in engineering systems.
- Apply the second law to closed and open systems and assess device efficiency using Carnot cycles.
- Analyse engineering systems such as heat engines, refrigerators and heat pumps from a second-law perspective.

Suggested textbook pages: Chapter 6 (approx. pages 156-200)